

Dabin Park

Ph.D. Student, School of Mathematics, Kyungpook National University

Research Assistant, Mathematics of Tomography Lab

Daegu, Republic of Korea · Advisor: Prof. Sunghwan Moon

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RESEARCH INTERESTS

Inverse problems; integral geometry and tomography (photoacoustic tomography, Radon-type transforms); operator learning and scientific machine learning for partial differential equations; finite element methods.

EDUCATION

Kyungpook National University — Ph.D. student in Mathematics 2023.09 – present

Kyungpook National University — M.S. in Mathematics 2022.03 – 2023.08

- GPA 4.30 / 4.30

Kyungpook National University — B.S. in Mathematics, *Magna Cum Laude* 2018.03 – 2022.02

- GPA 3.81 / 4.30 (major 4.14 / 4.30)

RESEARCH EXPERIENCE

Mathematics of Tomography Lab, Kyungpook National University — Research Assistant 2022.03 – present

- Advisor: Prof. Sunghwan Moon.
- Inverse problems for the wave and fractional wave equations; deep learning for photoacoustic tomography and operator learning for parametric PDEs.

Florida State University, Tallahassee, FL, USA — Visiting Researcher 2024.04 – 2025.07

- BK21 FOUR Graduate Student Overseas Training Program: 12-month fellowship, extended by a further 3 months at own expense. Host: Prof. Sanghyun Lee.
- Inverse and multiscale problems for physical equations; machine learning and finite element methods for inverse elliptic PDEs.

PUBLICATIONS

Peer-reviewed journal articles

1. **D. Park**, S. Lee, S. Moon. *A machine learning and finite element framework for inverse elliptic PDEs via Dirichlet-to-Neumann mapping*. Journal of Computational and Applied Mathematics, **486** (2026), 117679. [doi:10.1016/j.cam.2026.117679](https://doi.org/10.1016/j.cam.2026.117679)
2. G. Hwang, G. Jeon, S. Moon, **D. Park**. *Implicit learning to determine variable sound speed and the reconstruction operator in photoacoustic tomography*. Applied Mathematics in Science and Engineering, **34**(1) (2026), 2661458. [doi:10.1080/27690911.2026.2661458](https://doi.org/10.1080/27690911.2026.2661458)
3. **D. Park**, S. Moon. *Reconstruction of an initial function from the solutions of the fractional wave equation on the light cone trace*. Mathematical Methods in the Applied Sciences, **47**(12) (2024), 9469–9475. [doi:10.1002/mma.10078](https://doi.org/10.1002/mma.10078)

Refereed conference proceedings

4. F. Saeed, **D. Park**, S. S. A. Shah, J. Lee, A. Rehman, J.-M. Kang, I.-M. Kim, S. Yun, S. Moon. *Secure and privacy-preserving intrusion detection in wireless sensor networks via perturbation adversarial training*. Proceedings of the 41st ACM/SIGAPP Symposium on Applied Computing (SAC '26), Thessaloniki, Greece (2026), 2081–2087. [doi:10.1145/3748522.3779960](https://doi.org/10.1145/3748522.3779960)

In preparation

5. **D. Park**, S. Lee, S. Moon. *T-FEONet: A Transformer-based finite element operator network for time-dependent parametric PDEs*.

AWARDS

Top Prize , Graduate Student Paper Presentation Contest, Fall Conference of the Korean Data and Information Science Society	2023.11
Participation Prize , Annual Meeting and Research Presentation of the Youngnam Mathematical Society	2023.07
Award for Presentation Excellence , 15th Graduate Student Workshop on Mathematics	2022.07
Magna Cum Laude on B.S. graduation, Kyungpook National University	2022.02

CONFERENCE PRESENTATIONS

Secure and privacy-preserving intrusion detection in wireless sensor networks via perturbation adversarial training

- **Oral**. 41st ACM/SIGAPP Symposium on Applied Computing (SAC '26), Grand Hotel Palace, Thessaloniki, Greece, 2026.03.

A machine learning and finite element framework for inverse elliptic PDEs via Dirichlet-to-Neumann mapping

- **Oral (contributed student talk)**. 2025 Southeast Applied and Computational Math Student Workshop, Auburn University, Auburn, AL, USA, 2025.04.06.

Implicit learning for the variable sound speed and the reconstruction operator in photoacoustic tomography

- **Oral**. Fall Conference of the Korean Data & Information Science Society, Daegu University, 2023.11.03–11.04. Top Prize

Reconstruction of the initial functions from the solutions of the fractional wave equation on the light cone trace

- **Poster**. Annual Meeting of the Youngnam Mathematical Society, Kyungpook National University, Daegu, 2023.07.14–07.16. Participation Prize
- **Oral (in English)**. 15th Graduate Student Workshop, online, 2022.07.26–07.28. *Award for Presentation Excellence*
- **Oral**. Youngnam Mathematical Society, Dong-A University, Busan, 2022.06.23–06.24.
- **Poster**. KSIAM Spring Conference, Institute for Basic Science, 2022.05.27–05.29.
- **Oral**. Harmonic Analysis Workshop, Sungkyunkwan University, online, 2022.02.16–02.18.

SERVICE

Organizing Committee, Workshop for Young Mathematicians in Korea (WYMK) 2026

CONFERENCES AND WORKSHOPS ATTENDED

2026 KMS Yeongnam Branch AI Summer School, Pusan National University, Busan	2026.08.03 – 08.06
Yeungnam Mathematical Society, Busan	2026.07
KSIAM Annual Meeting	2025.11
SIAM Conference on Computational Science and Engineering (CSE25), Fort Worth, TX, USA	2025.03.03 – 03.07
ICIAM 2023, Waseda University, Tokyo, Japan	2023.08.20 – 08.25
KMS Spring Meeting, Daejeon Convention Center	2023.04.27 – 04.29
Winter AI Tutorial 2023 — Graph Neural Networks and Generative Models	2023.02.21 – 02.22
KSIAM Fall Conference, Sonocalm Jeju	2022.11.24 – 11.27
Understanding Graphs and Building GNNs, Coredottoday training course	2022.08
KSIAM Fall Conference, BEXCO, Busan	2021.12.02 – 12.05

COMPETITIONS

Nongnet Agricultural Price Fluctuation Forecasting AI Competition 2022.09

- Predicting the rate of change of agricultural prices against a reference date.
- **18th place** out of 69 teams. [Competition page](#)

SKILLS

- **Scientific computing:** FEniCS / legacy dolfin, finite element methods
- **Machine learning:** PyTorch (CUDA), operator learning, Transformer architectures
- **Languages and tools:** Python, Julia, MATLAB, ~~ETX~~ $\text{\LaTeX}$, Git, Docker / Dev Containers, Linux (WSL2)